

Introduction to Static Optimization

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Outline

- ① Formulation of the Programming Problem
- ② Alternative Formulations of the Problem
- ③ Kuhn-Tucker Theorem
- ④ Illustration through a Consumer-Choice Problem
- ⑤ MATLAB fmincon/Octave sqp

- ⑥ Additional Examples (Optional)
 - ① Intertemporal Consumption Problem
 - ★ A two-period problem as an introduction to dynamic optimization
 - ② Nash Bargaining
 - ★ A non-linear constraint & as an application (labor, trade, etc)

Formulation of the Programming Problem

Problem 1

- Let $f, g_1, g_2, \dots, g_n : \mathbb{R}^m \rightarrow \mathbb{R}$ be functions.
- Solve:

$$\begin{aligned} & \max_{x \in \mathbb{R}^m} f(x) \\ & \text{subject to } g_j(x) \geq 0 \text{ for each } j \in \{1, 2, \dots, n\}. \end{aligned}$$

- The problem is to find a point x of $\mathbb{R}^n$, subject to $g_j(x) \geq 0$ for all $j \in \{1, 2, \dots, n\}$, and such that $f(x)$ is a maximum.

Formulation of the Programming Problem

Remark 12.1

In Problem 1:

- ① ① The inequalities $g_j(x) \geq 0$ are the *constraints* of the program/problem.
- ② ② The function f is the *objective function* of the program.
- ③ ③ The set D of points which satisfy the constraints,

$$D := \{x \in \mathbb{R}^m \mid g_j(x) \geq 0 \text{ for each } j \in \{1, 2, \dots, n\}\},$$

is the *domain* of the program.

- ④ ④ The program is *bounded* if $f(x)$ is bounded above in the domain.
- ② If a condition of the form $g(x) \leq 0$ is imposed on x , it is enough to write $-g(x) \geq 0$.
- ③ If it is required to impose a condition on x of the form $g(x) = 0$, then it is enough to write $g(x) \geq 0$ and $-g(x) \geq 0$.

Binding Constraints

Definition

- The inequality constraint $g_j(x) \geq 0$ is *binding* at $x^* \in D$ if $g_j(x^*) = 0$.
- The inequality constraint $g_j(x) \geq 0$ is *slack* (not binding) $x^* \in D$ if $g_j(x^*) > 0$.

Which Constraints are Binding?

- We'll study how to solve constrained maximization (minimization) problems (analytically and numerically).
- Before taking derivatives or setting up a Lagrangian, the first step would be to figure out which constraints are binding at an optimum.
- Why?
 - ▶ To understand the main (economic) trade-off in the problem.
 - ▶ To simplify the problem and thus the resulting analysis.

Which Constraints are Binding?

Examples

- ① Non-negativity constraints (or something alike)?
 - ▶ E.g. Non-negative consumption $c \geq 0$.
 - ★ Under $\lim_{c \rightarrow 0} u'(c) = \infty$, an interior solution $c^* > 0$ is much easier to find!
 - ★ First-order equality condition vs inequality condition
 - ▶ In contrast, if a non-negativity condition (e.g. a borrowing constraint, a short-sale constraint, a limited liability constraint, the zero lower bound, etc) is binding, that may (drastically) change the solution. (And solving the first-order condition might not be helpful.)
- ② Which IC (incentive-compatibility) constraints are binding?
 - ▶ An agency problem might involve a lot of IC constraints. Understanding which IC constraints are binding leads to a better understanding of the problem, and the resulting analysis would be much less complicated.
- ③ For an equality constraint, which side of inequality constraints is binding?
 - ▶ A better understanding of the problem but also it gives us the “correct” sign of the Lagrange multiplier.
 - ▶ A budget constraint: $p_1x_1 + p_2x_2 \leq b$ instead of the opposite.

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Alternative Formulations of the Programming Problem

Overview (No worries about the details at this point)

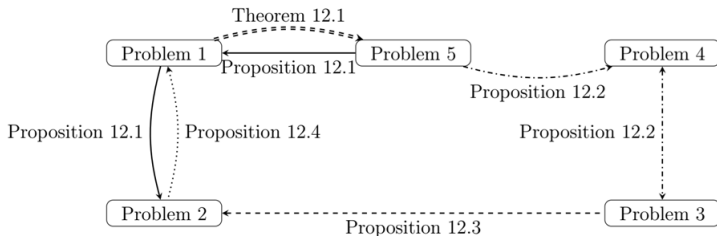


Figure 12.4: Relationship among Problems 1 to 5

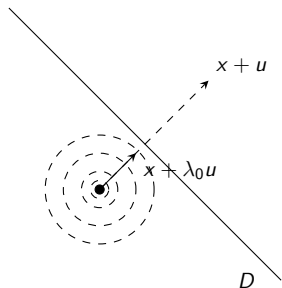
- Solid arrow: w/o any assumption
- Dash-dotted arrow: $f, g_1, g_2, \dots, g_n$ all differentiable
- Dashed arrow: $f, g_1, g_2, \dots, g_n$ all differentiable + f concave
- Dotted arrow: $f, g_1, g_2, \dots, g_n$ all concave
- Double-dashed arrow: $f, g_1, g_2, \dots, g_n$ all concave + Assumptions of Theorem 12.1

Alternative Formulations of the Programming Problem

Problem 2 (Local Maximum Problem)

Find $x \in D$ which possess the following property: for every point $u \in \mathbb{R}^m$, there exists $\lambda_0 > 0$ such that if $0 \leq \lambda \leq \lambda_0$ and $x + \lambda u \in D$ then $f(x + \lambda u) \leq f(x)$.

- Guess a solution x such that $f(x) \geq f(x + \lambda u)$ for any small local deviation $x + \lambda u$
- Assuming the solution x , the necessary condition is $\lambda = 0$ (λ is a scalar; useful when x is a vector or a function)



- Macro: guess $(c_t^*)_t$, increasing consumption today and decreasing consumption tomorrow
- Micro: guess a strategy profile, deviating for exactly one period to see if cheating is profitable (one-shot deviation principle)

Alternative Formulations of the Programming Problem

Problem 3

Suppose that $f, g_1, g_2, \dots, g_n$ are differentiable.

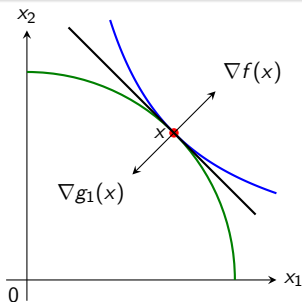
Find $x \in D$ such that

$$\langle \nabla f(x), u \rangle \leq 0$$

(the angle between $\nabla f(x)$ and u is at least as large as 90°) for every $u \in \mathbb{R}^m$ for which

$$g_j(x) = 0 \text{ implies } \langle \nabla g_j(x), u \rangle \geq 0.$$

(the angle between $\nabla g_j(x)$ and u is at least as small as 90°)



- When $n = 1$ (one constraint):
- $\nabla f(x)$ and $\nabla g_1(x)$ are parallel and in the opposite direction:

$$\nabla f(x) + y_1 \nabla g_1(x) = \mathbf{0}.$$

Alternative Formulations of the Programming Problem

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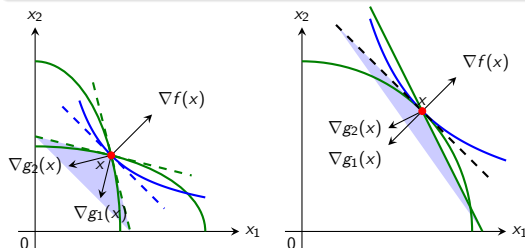
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$$g_j(x) = 0 \text{ implies } \langle \nabla g_j(x), u \rangle \geq 0.$$

(the angle between $\nabla g_j(x)$ and u is at least as small as 90°)



- It is not possible to increase the objective without violating at least one binding constraint.
- There is no feasible direction at x along which the objective increases.

Alternative Formulations of the Programming Problem

Problem 4 (Lagrange's Problem)

Suppose that $f, g_1, g_2, \dots, g_n$ are differentiable.

Find $\bar{x} \in D$ and $\bar{y} = (\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n) \in \mathbb{R}_+^n$ such that
 $\bar{y}_j \geq 0$ for every j

$$\frac{\partial f}{\partial x_i}(\bar{x}) + \sum_{j=1}^n \bar{y}_j \frac{\partial g_j}{\partial x_i}(\bar{x}) = 0 \text{ for all } i \in \{1, 2, \dots, m\}; \text{ and} \quad (\text{FOC})$$

$$\sum_{j=1}^n \bar{y}_j g_j(\bar{x}) = 0. \quad (\text{Complementary Slackness})$$

$$\underbrace{F(x, y)}_{\text{Lagrangian}} := \underbrace{f(x)}_{\text{objective}} + \sum_{j=1}^n \underbrace{y_j}_{\text{Lagrange multiplier}} \underbrace{g_j(x)}_{g_j(x) \geq 0}.$$

Alternative Formulations of the Programming Problem

- FOC: Fixing $\bar{y}$, the FOCs for $F(\cdot, \bar{y})$ with respect to each x_j .

$$\frac{\partial f}{\partial x_i}(x) + \sum_{j=1}^n \bar{y}_j \frac{\partial g_j}{\partial x_i}(x) = 0 \text{ for all } i \in \{1, 2, \dots, m\}$$

- What is $\bar{y}_j$?
 - 1 The marginal valuation of the problem if the constraint j is infinitesimally relaxed.
 - 2 The marginal gain from infinitesimally violating constraint j (the “(shadow) price” /penalty you’d have to charge to deter that violation).

Alternative Formulations of the Programming Problem

- Complementary slackness:

$$\sum_{j=1}^n \bar{y}_j g_j(\bar{x}) = 0.$$

- If the j -th constraint is binding, then $g_j(\bar{x}) = 0$.
- If not, the marginal valuation is zero: $\bar{y}_j = 0$.
- So, either $g_j(\bar{x}) = 0$ (binding) or $\bar{y}_j = 0$ (slack):

$$\bar{y}_j g_j(\bar{x}) = 0 \text{ for all } j \in \{1, \dots, n\}.$$

- Since $\bar{y}_j \geq 0$ and $g_j(\bar{x}) \geq 0$ (as $\bar{x} \in D$), this is equivalent to:

$$\sum_{j=1}^n \bar{y}_j g_j(\bar{x}) = 0.$$

- Implication: $\bar{y}$ minimizes $F(\bar{x}, y)$.

$$F(\bar{x}, \bar{y}) = f(\bar{x}) + \underbrace{\sum_{j=1}^n \bar{y}_j g_j(\bar{x})}_{=0} \leq f(\bar{x}) + \sum_{j=1}^m \underbrace{y_j}_{\geq 0} \underbrace{g_j(\bar{x})}_{\geq 0} = F(\bar{x}, y).$$

Alternative Formulations of the Programming Problem

Problem 5

Find

$$\bar{x} = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_m) \in \mathbb{R}^m \text{ and } \bar{y} = (\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n) \in \mathbb{R}_+^n$$

such that the pair $(\bar{x}, \bar{y})$ is a saddle point of the function

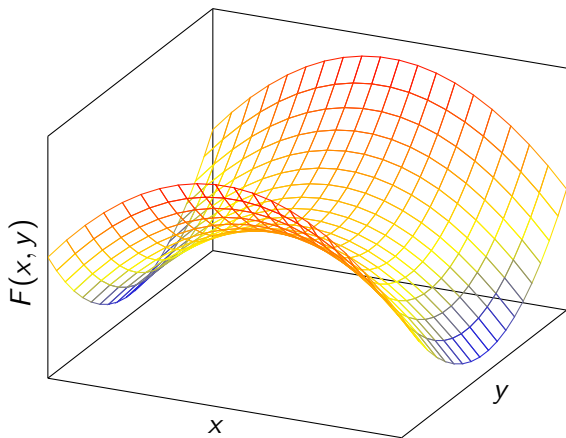
$$F(x, y) := f(x) + \sum_{j=1}^n y_j g_j(x).$$

That is, find $(\bar{x}, \bar{y})$ such that

$$\overbrace{F(x, \bar{y}) \leq F(\bar{x}, \bar{y})}^{\bar{x} \text{ maximizes } F(\cdot, \bar{y})} \leq \underbrace{F(\bar{x}, y)}_{\bar{y} \text{ minimizes } F(\bar{x}, \cdot)} \leq F(\bar{x}, y) \text{ for all } x \in \mathbb{R}^m \text{ and } y \in \mathbb{R}_+^n.$$

We call the function $F : \mathbb{R}^m \times \mathbb{R}_+^n \rightarrow \mathbb{R}$ the *Lagrangian function*.

Alternative Formulations of the Programming Problem



- Given $\bar{y}$, $\max_x F_{x, \bar{y}}$.
- Given $\bar{x}$, $\min_y F(\bar{x}, y)$.

Kuhn-Tucker Theorem (Abstract, Optional Exercise)

Proposition 12.1

- ① If $(\bar{x}, \bar{y}) \in \mathbb{R}^m \times \mathbb{R}_+^n$ is a solution of Problem 5, then $\bar{x}$ is a solution of Problem 1.
 - ② If $x \in D$ is a solution of Problem 1, then it is a solution of Problem 2.
-
- ①
 - ▶ Given $\bar{x}$, $F(\bar{x}, \bar{y}) \leq F(\bar{x}, y)$ for all $y \in \mathbb{R}_+^n \Rightarrow \bar{x} \in D$
(what would happen if $\bar{x} \notin D$? Can we minimize $F(\bar{x}, y)$?).
 - ▶ Given $\bar{y}$, $F(x, \bar{y}) \leq F(\bar{x}, \bar{y})$ for all $x \in D \Rightarrow f(x) \leq f(\bar{x})$ for all $x \in D$.
 - ② A global maximizer $\Rightarrow$ a local maximizer.

Exercise 12.3

Let $(\bar{x}, \bar{y})$ be a solution of Problem 5. Show that

$$\bar{y}_j g_j(\bar{x}) = 0 \text{ for each } j \in \{1, 2, \dots, n\}.$$

- If $\bar{y}_j > 0$, then $g_j(\bar{x}) = 0$.
Otherwise (if $g_j(\bar{x}) > 0$), you can decrease $\bar{y}_j$ a bit, a contradiction.

Kuhn-Tucker Theorem

Theorem 12.1 (Kuhn-Tucker Theorem)

Suppose that the functions $f, g_1, g_2, \dots, g_n$ are **concave**.

Suppose further that there exists at least one point $x \in \mathbb{R}^m$ with the following two properties: (i) $g_j(x) \geq 0$ for all $j \in \{1, 2, \dots, n\}$ (i.e., the constraints are satisfied); and (ii) $g_j(x) \neq 0$ for all the functions which are not linear affine.

- 1 Then, Problems 1, 2, and 5 are equivalent.
- 2 Moreover, if $f, g_1, g_2, \dots, g_n$ are differentiable, then **all the five problems are equivalent**.

Illustration through a Consumer-Choice Problem

- As an example, consider:

$$f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha} \text{ and} \\ g_1(x_1, x_2) = m - p_1 x_1 - p_2 x_2.$$

- The constraint $g_1(x_1, x_2) \geq 0$ is binding (why?).
- The additional non-negativity constraints

$$g_2(x_1, x_2) = x_1 \geq 0 \text{ and } g_3(x_1, x_2) = x_2 \geq 0$$

are not binding (why?).

- Note: Since the constraint is a compact set and the objective function is a continuous strictly concave function, the solution (x_1^*, x_2^*) exists uniquely.

Illustration through a Consumer-Choice Problem

- First, a (familiar) economic intuition tells us that at an optimal choice (x_1^*, x_2^*) , the indifference curve (or the level curve)

$$\{(x_1, x_2) \in \mathbb{R}^2 \mid f(x_1, x_2) = f(x_1^*, x_2^*)\}$$

is tangent to the budget constraint

$$\{(x_1, x_2) \in \mathbb{R}^2 \mid g(x_1, x_2) = 0\}$$

(otherwise, it is always better for the consumer to slightly change her consumption bundle).

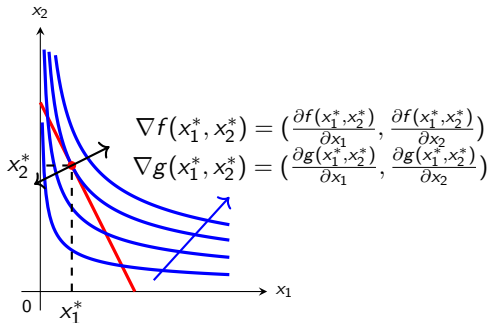
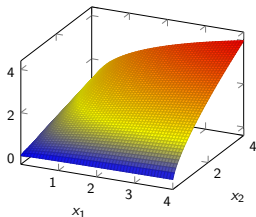


Illustration through a Consumer-Choice Problem

- Second, the gradient

$$\nabla f(x_1^*, x_2^*) = \left(\frac{\partial f(x_1^*, x_2^*)}{\partial x_1}, \frac{\partial f(x_1^*, x_2^*)}{\partial x_2} \right)$$

is perpendicular to the indifference curve $f(x_1, x_2) = f(x_1^*, x_2^*)$.¹

- Likewise, the gradient

$$\nabla g(x_1^*, x_2^*) = \left(\frac{\partial g(x_1^*, x_2^*)}{\partial x_1}, \frac{\partial g(x_1^*, x_2^*)}{\partial x_2} \right)$$

is perpendicular to the budget constraint $g(x_1, x_2) = 0$.

¹Intuitively, by differentiability,

$$0 = f(x_1, x_2) - f(x_1^*, x_2^*) \approx \begin{pmatrix} \frac{\partial f(x_1^*, x_2^*)}{\partial x_1} \\ \frac{\partial f(x_1^*, x_2^*)}{\partial x_2} \end{pmatrix} \cdot \begin{pmatrix} x_1 - x_1^* \\ x_2 - x_2^* \end{pmatrix}.$$

Illustration through a Consumer-Choice Problem

- At the optimal solution (x_1^*, x_2^*) , it is necessary (and in this case also sufficient) that the gradients $\nabla f(x_1^*, x_2^*)$ and $\nabla g(x_1^*, x_2^*)$ are parallel.
- In other words, there is some number $y_1 \in \mathbb{R}_+$ such that

$$\nabla f(x_1^*, x_2^*) + y_1 \nabla g_1(x_1^*, x_2^*) = 0.$$

- The non-negative value y_1 is the marginal valuation of the constraint g_1 .
- It tells the rate at which the maximum value of the objective function changes with respect to a marginal relaxation of the constraint.
 - ▶ If $g(x_1^*, x_2^*) > 0$ then $y_1 = 0$.
 - ▶ If $g(x_1^*, x_2^*) = 0$ then $y_1 \geq 0$.

Illustration through a Consumer-Choice Problem

- Thus, the maximization problem reduces to the following problem.
First, set up the Lagrangian function

$$L(x_1, x_2, y_1) := f(x_1, x_2) + y_1 g_1(x_1, x_2).$$

- Then, find (x_1^*, x_2^*) and $y_1^* \in \mathbb{R}_+$ such that

$$\frac{\partial}{\partial x_i} L(x_1^*, x_2^*, y_1^*) = 0 \text{ for each } i \in \{1, 2\}, \text{ and} \quad (\text{FOC})$$

$$y_1^* g_1(x_1^*, x_2^*) = 0. \quad (\text{Complementary Slackness})$$

Illustration through a Consumer-Choice Problem

- The first-order conditions

$$\begin{aligned}\frac{\partial f(x_1^*, x_2^*)}{\partial x_1} &= y_1 p_1 \text{ and} \\ \frac{\partial f(x_1^*, x_2^*)}{\partial x_2} &= y_1 p_2\end{aligned}$$

yield:

$$\frac{x_2^*}{x_1^*} = \frac{1 - \alpha}{\alpha} \frac{p_1}{p_2}.$$

- Together with the budget constraint (which is binding)

$$p_1 x_1^* + p_2 x_2^* = m,$$

we obtain

$$(x_1^*, x_2^*) = \left(\frac{\alpha m}{p_1}, \frac{(1 - \alpha)m}{p_2} \right).$$

Illustration through a Consumer-Choice Problem

- Then, it follows from one of the first-order conditions

$$y_1^* = \frac{\frac{\partial f(x_1^*, x_2^*)}{\partial x_1}}{p_1}$$

that

$$y_1^* = \frac{\alpha(x_1^*)^{\alpha-1}(x_2^*)^{1-\alpha}}{p_1} = \frac{\alpha(x_1^*)^\alpha(x_2^*)^{1-\alpha}}{p_1 x_1} = \frac{(x_1^*)^\alpha(x_2^*)^{1-\alpha}}{m}.$$

- Similarly, it follows from the other first-order condition

$$y_1^* = \frac{\frac{\partial f(x_1^*, x_2^*)}{\partial x_2}}{p_2}$$

that

$$y_1^* = \frac{(1-\alpha)(x_1^*)^\alpha(x_2^*)^{-\alpha}}{p_2} = \frac{(1-\alpha)(x_1^*)^\alpha(x_2^*)^{1-\alpha}}{p_2 x_2} = \frac{(x_1^*)^\alpha(x_2^*)^{1-\alpha}}{m}.$$

Illustration through a Consumer-Choice Problem

- The Lagrange multiplier y_1^* is the marginal valuation of the problem if the constraint $g_1(x_1, x_2) \geq 0$ is infinitesimally relaxed.
- If m is infinitesimally increased, then

$$\begin{aligned}\frac{d}{dm}L(x_1^*, x_2^*, y_1^*) &= \frac{d}{dm} \left(\underbrace{f(x_1^*(m), x_2^*(m)) + y_1^*(m)(m - p_1 x_1^*(m) - p_2 x_2^*(m))}_{f(x_1^*, x_2^*) + y_1^*(m - p_1 x_1^* - p_2 x_2^*)} \right) \\&= \underbrace{\left(\frac{\partial f(x_1^*, x_2^*)}{\partial x_1} - y_1^* p_1 \right)}_{=0: \text{ FOC}} \frac{dx_1}{dm} + \underbrace{\left(\frac{\partial f(x_1^*, x_2^*)}{\partial x_2} - y_1^* p_2 \right)}_{=0: \text{ FOC}} \frac{dx_2}{dm} \\&\quad + \underbrace{(m - p_1 x_1^* - p_2 x_2^*)}_{=0: \text{ Constraint}} \frac{dy_1}{dm} + y_1^* \underbrace{\frac{dm}{dm}}_{=1} \\&= y_1^*.\end{aligned}$$

- Envelope Theorem: Only the *direct* effect of m on the Lagrangian survives, as the indirect effects through $x_1^*(m)$ and $x_2^*(m)$ vanish.

Illustration through a Consumer-Choice Problem

- In our consumer choice problem, recalling

$$(x_1^*, x_2^*) = \left(\frac{\alpha m}{p_1}, \frac{(1 - \alpha)m}{p_2} \right),$$

we can express y_1^* as follows:

$$\begin{aligned} \frac{d}{dm} L(x_1^*, x_2^*, y_1^*) &= \frac{d}{dm} \left(f(x_1^*, x_2^*) + y_1^* \underbrace{(m - p_1 x_1^* - p_2 x_2^*)}_{=0} \right) \\ &= \frac{d}{dm} f(x_1^*, x_2^*) \\ &= \frac{\partial}{\partial x_1} f(x_1^*, x_2^*) \frac{dx_1}{dm} + \frac{\partial}{\partial x_2} f(x_1^*, x_2^*) \frac{dx_2}{dm} \\ &= \alpha (x_1^*)^{\alpha-1} (x_2^*)^{1-\alpha} \frac{x_1^*}{m} + (1 - \alpha) (x_1^*)^{\alpha} (x_2^*)^{-\alpha} \frac{x_2^*}{m} \\ &= \frac{(x_1^*)^{\alpha} (x_2^*)^{1-\alpha}}{m}. \end{aligned}$$

Illustration through a Consumer-Choice Problem

- Also, y_1^* minimizes $F(x_1^*, x_2^*, y_1)$.

- If $y_1^* = 0$ then

$$\begin{aligned} F(x_1^*, x_2^*, y_1) &= f(x_1^*, x_2^*) + \underbrace{y_1}_{\geq 0} \underbrace{g(x_1^*, x_2^*)}_{\geq 0} \\ &\geq f(x_1^*, x_2^*) + \underbrace{y_1^*}_{=0} \underbrace{g(x_1^*, x_2^*)}_{\geq 0} = F(x_1^*, x_2^*, y_1^*). \end{aligned}$$

- If $y_1^* > 0$ then $g(x_1^*, x_2^*) = 0$. Then,

$$\begin{aligned} F(x_1^*, x_2^*, y_1) &= f(x_1^*, x_2^*) + \underbrace{y_1}_{\geq 0} \underbrace{g(x_1^*, x_2^*)}_{=0} \\ &= f(x_1^*, x_2^*) + \underbrace{y_1^*}_{>0} \underbrace{g(x_1^*, x_2^*)}_{=0} = F(x_1^*, x_2^*, y_1^*). \end{aligned}$$

Illustration through a Consumer-Choice Problem

- Suppose that (x_1^*, x_2^*) is an optimal consumption bundle.
- For ease of exposition, we consider the maximization of

$$\log(x_1^\alpha x_2^{1-\alpha}) = \alpha \log x_1 + (1 - \alpha) \log x_2.$$

- To respect the budget constraint, consider

$$(x_1, x_2) = (x_1^* + \Delta x_1, x_2^* - \frac{p_1}{p_2} \Delta x_1).$$

- The utility from (x_1, x_2) is

$$\alpha \log(x_1^* + \Delta x_1) + (1 - \alpha) \log(x_2^* - \frac{p_1}{p_2} \Delta x_1).$$

Illustration through a Consumer-Choice Problem

- The utility from (x_1, x_2) is

$$\alpha \log(x_1^* + \Delta x_1) + (1 - \alpha) \log(x_2^* - \frac{p_1}{p_2} \Delta x_1).$$

- Since it is optimal to set $\Delta x_1 = 0$, the first-order condition with respect to Δx_1 at $\Delta x_1 = 0$ is:

$$0 = \alpha \frac{1}{x_1^* + \Delta x_1} - (1 - \alpha) \frac{p_1}{p_2} \frac{1}{x_2^* - \frac{p_1}{p_2} \Delta x_1} \Bigg|_{\Delta x_1 = 0}.$$

- That is,

$$\alpha \frac{1}{x_1^*} = (1 - \alpha) \frac{p_1}{p_2} \frac{1}{x_2^*}.$$

- Rearranging,

$$\frac{x_2^*}{x_1^*} = \frac{1 - \alpha}{\alpha} \frac{p_1}{p_2}.$$

MATLAB Solver fmincon

MATLAB Solver **fmincon**

$$\begin{array}{ll}\min_{x \in \mathbb{R}^n} & f(x) \\ \text{subject to} & c(x) \leq \mathbf{0}, \quad (\text{non-linear inequality constraints}) \\ & \text{ceq}(x) = \mathbf{0}, \quad (\text{non-linear equality constraints}) \\ & Ax \leq b, \quad (\text{linear inequality constraints}) \\ & (\text{Aeq})x = \text{beq}, \quad (\text{linear equality constraints}) \\ & \text{lb} \leq x \leq \text{ub}. \quad (\text{lower/upper bounds})\end{array}$$

- Maximize $f \Leftrightarrow$ Minimize $-f$.

Octave Solver sqp

Sequential Quadratic Programming

Octave Solver sqp

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \text{phi}(x) \\ \text{subject to } & g(x) = \mathbf{0}, & (\text{non-linear equality constraints}) \\ & h(x) \geq \mathbf{0}, & (\text{non-linear inequality constraints}) \\ & \text{lb} \leq x \leq \text{ub}. & (\text{lower/upper bounds}) \end{aligned}$$

- Maximize $f \Leftrightarrow$ Minimize $-f$.
- (Octave fmincon, which requires the optim package, is a wrapper that uses Octave sqp.)

Python Solver `scipy.optimize.minimize`

Optional (For your information)

`method='trust-constr'`

$$\min_{x \in \mathbb{R}^n} f(x)$$

subject to $\text{lb}_c \leq c(x) \leq \text{ub}_c$, (linear/non-linear constraints)

$\text{lb} \leq x \leq \text{ub}$. (bounds)

- Maximize $f \Leftrightarrow$ Minimize $-f$.
- Linear constraints: $c(x) = Ax$.

Consumer Choice Example

- Use fmincon/sqp to solve the following consumer choice problem:

$$\max_{(x_1, x_2) \in \mathbb{R}_+^2} x_1^\alpha x_2^{1-\alpha}$$

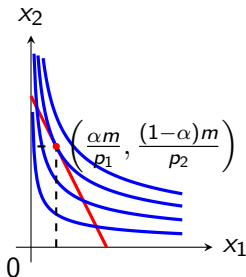
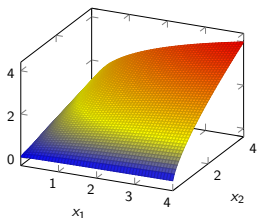
$$\text{subject to } p_1 x_1 + p_2 x_2 \leq m \Leftrightarrow (p_1 \quad p_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq m.$$

- Letting the Lagrangian

$$L(x_1, x_2) = x_1^\alpha x_2^{1-\alpha} + \lambda(m - p_1 x_1 - p_2 x_2),$$

the solution is

$$(x_1, x_2, \lambda) = \left(\frac{\alpha m}{p_1}, \frac{(1-\alpha)m}{p_2}, \frac{x_1^\alpha x_2^{1-\alpha}}{m} \right).$$



Intertemporal Utility Maximization Example (Optional)

- Intertemporal Utility Maximization Example:

$$\begin{aligned} \max_{(c_0, c_1) \in \mathbb{R}_+^2} \quad & u(c_0) + \beta u(c_1) \\ \text{subject to} \quad & (a_0 + y_0) + \frac{y_1}{r_0} \geq c_0 + \frac{c_1}{r_0} \end{aligned}$$

- Letting the Lagrangian

$$L(x_1, x_2) = u(c_0) + \beta u(c_1) + \lambda \left((a_0 + y_0) + \frac{y_1}{r_0} - c_0 - \frac{c_1}{r_0} \right),$$

it can be seen that the solution (c_0, c_1) satisfies the Euler equation

$$u'(c_0) = \beta r_0 u'(c_1) (= \lambda).$$

When $u(c) = \log(c)$, it turns out that

$$(c_0, c_1) = \left(\frac{1}{1+\beta} A, \frac{\beta r_0}{1+\beta} A \right), \text{ where } A = a_0 + y_0 + \frac{y_1}{r_0}.$$

Intertemporal Utility Maximization Example

- Suppose that (c_0^*, c_1^*) solves the problem.
- Consider an alternative consumption plan that respects the budget constraint:

$$(c_0, c_1) = (c_0^* + \Delta c_0, c_1^* - r_0 \Delta c_0).$$

- The utility from the new consumption plan is:

$$u(c_0^* + \Delta c_0) + \beta u(c_1^* - r_0 \Delta c_0).$$

- Since it is optimal to set $\Delta c_0 = 0$, the first-order condition with respect to Δc_0 at $\Delta c_0 = 0$ is:

$$0 = u'(c_0^* + \Delta c_0) - r_0 \beta u'(c_1^* - r_0 \Delta c_0) \Big|_{\Delta c_0=0},$$

that is,

$$u'(c_0^*) = \beta r_0 u'(c_1^*).$$

Intertemporal Utility Maximization Example

Lagrangian (Optional)

- Let

$$\mathcal{L} = u(c_0) + \beta u(c_1) + \lambda \left(a_0 + y_0 + \frac{y_1}{r_0} - c_0 - \frac{c_1}{r_0} \right).$$

- The first-order conditions are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_0} &= u'(c_0) - \lambda = 0 \Rightarrow \lambda = u'(c_0); \text{ and} \\ \frac{\partial \mathcal{L}}{\partial c_1} &= \beta u'(c_1) - \frac{\lambda}{r_0} = 0 \Rightarrow \lambda = \beta r_0 u'(c_1). \end{aligned}$$

- Then, we obtain the Euler equation:

$$u'(c_0) = \beta r_0 u'(c_1).$$

- When $u(\cdot) = \log(\cdot)$, the Euler equation reduces to:

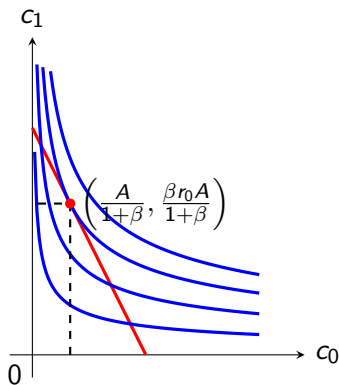
$$c_1 = \beta r_0 c_0.$$

Intertemporal Utility Maximization Example

Lagrangian

- Since the budget constraint is binding (why?), substituting the Euler equation into the budget constraint yields

$$(c_0, c_1) = \left(\frac{A}{1+\beta}, \frac{\beta r_0 A}{1+\beta} \right), \text{ where } A = a_0 + y_0 + \frac{y_1}{r_0}.$$



Nash Bargaining Example (Optional)

Definition 11.1

An (axiomatic two-player) bargaining problem is a pair $\langle U, d \rangle$, where:

- 1 U is a set of payoff profiles to agreements, which is a compact convex subset of $\mathbb{R}^2$;
- 2 $d \in U$ is a pair of payoff profiles to disagreement. Assume that there is $(u_1, u_2) \in U$ with $u_1 > d_1$ and $u_2 > d_2$.

Definition 11.2

The Nash bargaining solution assigns, to each bargaining problem $\langle U, d \rangle$, the pair of payoffs that solves the problem

$$\max_{(u_1, u_2) \in \mathbb{R}^2} (u_1 - d_1)(u_2 - d_2) \text{ subject to } (u_1, u_2) \in U \text{ and } (u_1, u_2) \geq (d_1, d_2).$$

Nash Bargaining Example

- Consider a bargaining problem $\langle U, d \rangle$, where

$$U := \{(u_1, u_2) \in \mathbb{R}^2 \mid u_1^2 + u_2^2 \leq 1\} \text{ and } d := (0, 0).$$

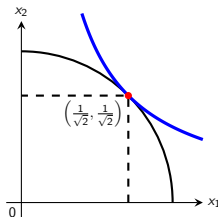
- The Nash bargaining solution of this problem is found by solving

$$\max u_1 u_2 \text{ subject to } u_1^2 + u_2^2 \leq 1 \text{ and } (u_1, u_2) \geq (0, 0).$$

- It can be seen that the solution is

$$(u_1, u_2) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

- The MATLAB Codes in the lecture notes



Nash Bargaining Example

Lagrangian

- While $u_1^2 + u_2^2 \leq 1$ is binding, $u_1 \geq 0$ and $u_2 \geq 0$ are not (why?).
- Let

$$\mathcal{L} = u_1 u_2 + \lambda(1 - u_1^2 - u_2^2).$$

- The first-order conditions are:

$$\frac{\partial \mathcal{L}}{\partial u_1} = u_2 - 2u_1\lambda = 0 \Rightarrow \lambda = \frac{u_2}{2u_1}; \text{ and}$$

$$\frac{\partial \mathcal{L}}{\partial u_2} = u_1 - 2u_2\lambda = 0 \Rightarrow \lambda = \frac{u_1}{2u_2}.$$

- Then, we obtain:

$$\frac{u_2}{2u_1} = \frac{u_1}{2u_2},$$

and thus

$$u_1^2 = u_2^2.$$

- Since $u_1 > 0$ and $u_2 > 0$, we have

$$u_1 = u_2.$$

Nash Bargaining Example

Lagrangian

- Since the constraint $u_1^2 + u_2^2 \leq 1$ is binding, we obtain:

$$(u_1, u_2) = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

- Also,

$$\lambda = \frac{1}{2}.$$